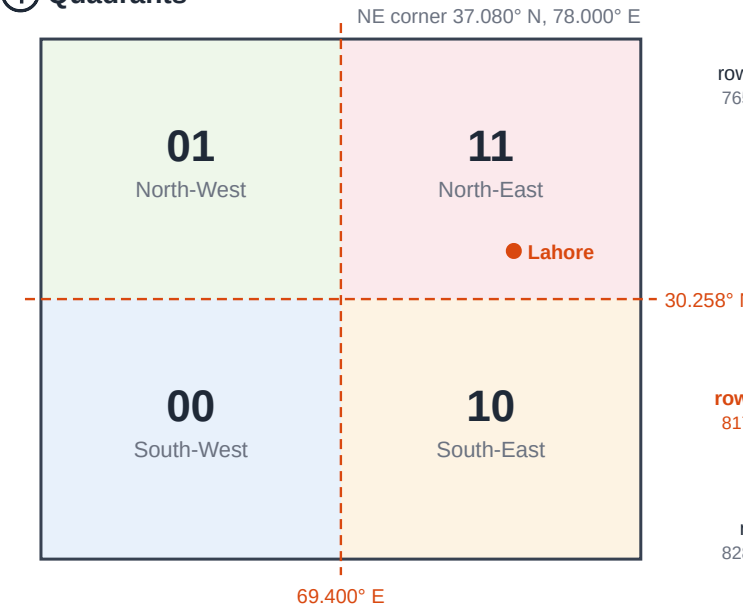
Pakistan National Grid — how a grid code is built

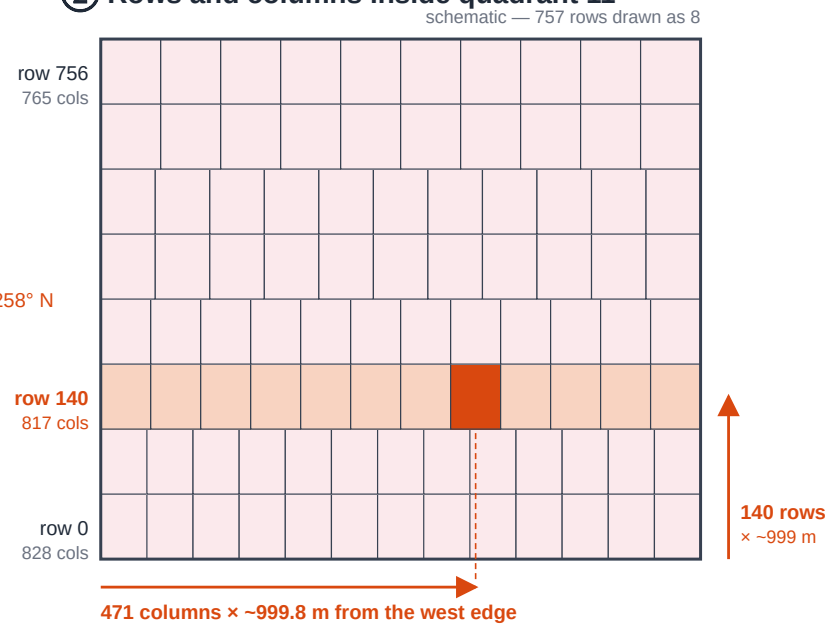
Worked example: central Lahore, 31.5204° N, 74.3587° E → 11-0140-0471-00

① Quadrants



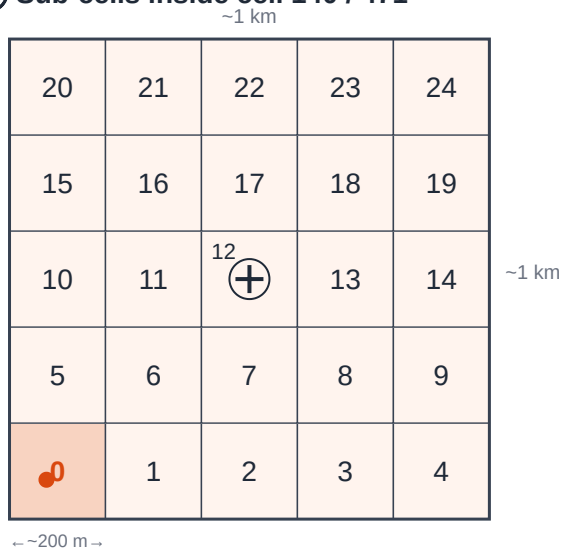
Is it north of 30.258° and east of 69.4°?
Yes and yes → quadrant 11 (number 3)

② Rows and columns inside quadrant 11



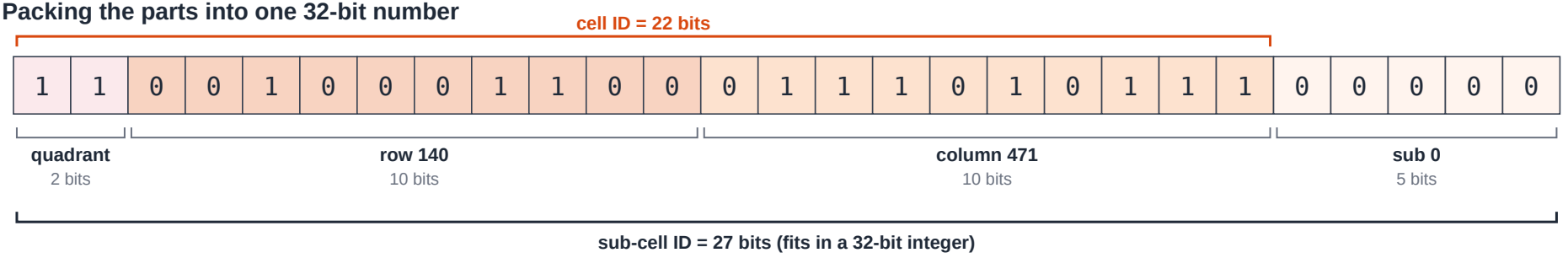
Rows: equal latitude bands, ~1 km tall.
Columns: each row gets its own count so cells stay ~1 km wide; rows further north are narrower, so they get fewer.

③ Sub-cells inside cell 140 / 471



● point: 0.08 up, 0.08 across the cell
sub-row 0, sub-col 0 → sub = 0 × 5 + 0 = 0
⊕ cell centroid (stored as a GPS point)
Numbering: sub = sub_row × 5 + sub_col (0–24)

④ Packing the parts into one 32-bit number



Row	`floor((31.5204 − 30.258) ÷ 0.0090119) = floor(140.08) = 140`
Column	`floor((74.3587 − 69.400) ÷ 0.0105263) = floor(471.08) = 471`
Cell	`(3 « 20) | (140 « 10) | 471 = 3,289,559`
Sub	`(3,289,559 « 5) | 0 = 105,265,888`
Parent	`105,265,888 » 5 = 3,289,559 (no lookup needed)`

Readable code

11-0140-0471-00

quadrant · row · column · sub